0. **Code:** vemuriu6191 vemuriu 6191.zip

1. Raise a polynomial/Program.cs

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```
namespace Raise_a_polynomial
{
    internal class Program
    {
        static int[] RaisePolynomialToPower(int[] coefficents, int power)
        {
            if (coefficents == null || power ≤ 0)
            {
                return null;
            }
            if (power == 1)
            {
                return coefficents;
            }
            int[] result = coefficents;
            for (int p = 2; p ≤ power; p++)
            {
                result = MultiplyPolynomials(result, coefficents);
            }
            return result;
        }
        static int[] MultiplyPolynomials(int[] poly1, int[] poly2)
        {
            int totalLength = poly1.Length + poly2.Length + 1;
            int[] result = new int[totalLength];
            for (int i = 0; i < poly1.Length; i++)
            {
                for (int j = 0; j < poly2.Length; j++)
                {
                    for (int k = 0; k < result.Length; k++)
```

B1

B2

ok

B6

B3

because the length-1 is repeat

$(length_1-1)+(length_2-1)+1 = length_1+length_2-1$

not needed!

only two-nested loops are needed!

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```
                {
                    result[i + j] += poly1[i] * poly2[j];
                }

            }
        }
        return result;
    }
    static void Main(string[] args)
    {
        int[] coefficents = new int[]
        {
            5,
            6,
            7,
            8,
            9
        };
        int n = 3;
        int[] result = RaisePolynomialToPower(coefficents, n);

        if (result ≠ null)
        {
            Console.WriteLine("The Polynomial raised to the power of " +
              n + " is:");
            for (int i = 0; i < result.Length; i++)
            {
                Console.WriteLine(result[i] + " ");
            }
        }
        else
```

B5

B4

13

1. **Raise a polynomial/Program.cs**

```
            {
                Console.WriteLine("The result is null");
            }
        }
    }
}
```